CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

ME342 HEAT AND MASS TRANSFER

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
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CHAMOS Matrusanstha Department of Mechanical Engineering

Me 342 Heat And Mass Trasfer	
CO1	Understand The Basic Laws Of Heat Transfer As Well As Ability To Understand And Solve Conduction, Convection And Radiation Problems.
CO2	Analyze Problems Involving Steady State Heat Conduction In Simple Geometries.
CO3	Understand The Fundamentals Of Convective Heat Transfer Process.
CO4	Ability To Design And Analyze The Performance Of Heat Exchangers Using LMTD And NTU Method.
CO5	Understand And Analyze Heat Transfer Through Extended Surfaces.
CO6	Examine Thermal Conductivity Of Different Geometries.
CO7	Analyze Unsteady State Heat Conduction Using Experimental Setup.

Sr no.	**Title Of Experiment (ME342)**	**Course Outcome**
1	Thermal Conductivity Of Composite Wall	**CO1, CO6**
2	Thermal Conductivity Of Insulating Powder	**CO1, CO6**
3	Thermal Conductivity Of Metal Rod	**CO1, CO6**
4	Unsteady State Heat Transfer Apparatus	**CO3,CO7**
5	Specific Heat of Air	**CO1,CO3**
6	Double Pipe Heat Exchanger	**CO4**
7	Heat Transfer From Pin Fin	**CO5**
8	Critical radius of insulation	**CO1, CO2**
9	Stephan Boltzmann law	**CO1**
10	Emissivity Measurement Apparatus	**CO1,CO3**

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Heat and Mass Transfer (ME342)

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Heat and Mass Transfer (ME303) **Date:**

EXPERIMENT NO 1
THERMAL CONDUCTIVITY OF COMPOSITE WALL

AIM:
1) To determine the Overall Thermal Conductivity of Composite Wall.
2) To check the Thermal Resistances of Composite Wall connected in series.

APPARATUS:
Thermal Conductivity Apparatus with Composite wall (Mild Steel, Asbestos & Plywood), C clamps.

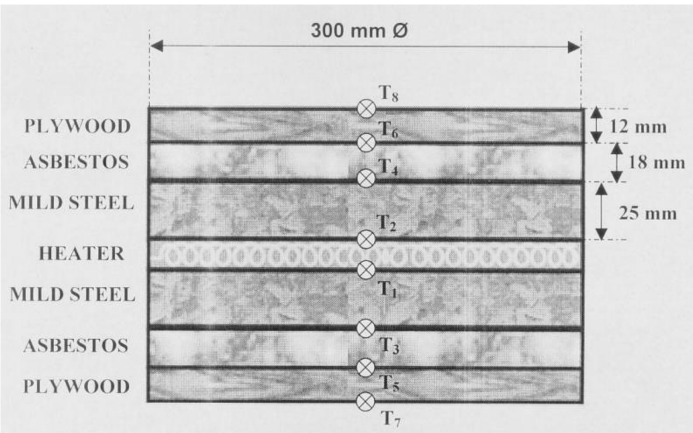
THEORY:
A sketch of the apparatus is shown in Figure. The essential parts are Heater plate, Composite Wall made of Mild Steel, Asbestos & Plywood and thermocouples in position as shown in the same figure.
For the measurement of the thermal conductivity K what is required is to have one dimensional heat flow through the flat specimen, an arrangement for maintaining its faces at the constant temperature and metering method to measure the heat flow through a known area.
Knowing the heat input to the central plate heater, the temperature difference across the each specimen, its thickness and the area, one can calculate the K by the following formula.

$$k = q * L / \left[2 * A * \left(T_h - T_c\right)\right]$$

Where,

k	Thermal Conductivity of the sample, W / m °C
q	Heat flow rate in the specimen, W
A	Area of the specimen, m^2
T_h	Hot side average temperature, °C
T_c	Cold side average temperature, °C
L	Thickness of the specimen, m

EXPERIMENTAL SET UP:



Thermal Conductivity of composite wall

PROCEDURE:

1. The specimens are placed in a chamber such that they cover the heating plate on either side, wherein MS is closer to the heating surface.
2. A predetermined heat input is given to the heater using the Dimmerstat.
3. After every 5 minutes the input of the heater (current & voltage) and the thermocouple readings are observed till a reasonably steady state condition is reached (steady temperature gradient).
4. More heat input is provided by Dimmerstat to take the next set of readings.

OBSERVATION TABLE:

Sr. No	**Volt (V)**	**Amp. (A)**	**T_1 (°C)**	**T_2 (°C)**	**T_3 (°C)**	**T_4 (°C)**	**T_5 (°C)**	**T_6 (°C)**	**T_7 (°C)**	**T_8 (°C)**

CALCULATIONS:

1) Heat Transfer Area Perpendicular to Heat Flow (m^2),

$$A = \left(\frac{\Pi}{4}\right) * D^2$$ Where, D = Diameter of plate = 0.3 m

Heat Input $Q = V * I$ Watts

2) Thermal Conductivity for individual specimen (W/ m °C),

$$k_i = \left[\frac{(Q * L_i)}{\left[2 * A_i * \left(T_{h_{i,av}} - T_{c_{i,av}}\right)\right]}\right]$$

Where,

i = 1, 2 and 3 for MS, Asbestos & Plywood Plate respectively.

$T_{h_{i,av}}$ = (T_1 + T_2) / 2 for MS plate,
(T_3 + T_4) / 2 for Asbestos & Plywood Plate,
(T_5 + T_6)/ 2 for Plywood Plate, °C

$T_{c_{i,av}}$ = (T_3 + T_4) / 2 for MS Plate,
(T_5 + T_6)/ 2 for Asbestos Plate,
(T_7 + T_8) / 2 for Plywood Plate, °C

L_i = Thickness of slab, m.

3) Overall Thermal Conductivity (W/ m °C)

$$K_{\text{overall}} = \frac{(Q * L_i)}{\left[2 * A * \left(\frac{(T_1 + T_2)}{2}\right) - \left(\frac{(T_7 + T_8)}{2}\right)\right]}$$

RESULT TABLE:

Sr. No.	Thermal Conductivity (W/ m °C)			
	M.S.	Asbestos	Wood	Overall
1				
2				
3				

CONCLUSION:

REVIEW QUESTION:

1. List the assumption made for the experiment.
2. Describe the set-up and explain experimental procedure.
3. Sketch the experiment set-up

Marks obtained:	Signature of faculty:	Date:

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Heat and Mass Transfer (ME303) **Date:**

EXPERIMENT NO 2
THERMAL CONDUCTIVITY OF INSULATING POWDER

AIM:
To determine the thermal conductivity of given insulating powder.

APPARATUS:
Conductivity Instrument consisting of Two Concentric Spheres, Insulating Powder and Control Panel with Voltmeter, Ampere meter, Digital Temperature Controller-cum-Indicator with selector switch, Dimerstat and Main On/ Off Switch.

THEORY:
Consider the transfer by heat conduction through the wall of a hollow sphere formed by the insulating powdered layer packed between two thin copper spheres.

Applying the Fourier's Law of heat conduction for a thin spherical layer of radius r and thickness dr, with temperature difference dT the heat transfer rate

$$q = -k * 4\prod r^2 * \frac{\partial T}{\partial r}$$

Where,

k Thermal Conductivity of the sample, W / m °C

Separating the variables and integrating in the limits r_i to r_o and T_i to T_o , we get

$$\left(\frac{q}{4\prod k}\right) * \left[\frac{1}{r_i} - \frac{1}{r_o}\right] = \left(T_i - T_o\right)$$

or,

$$q = \frac{4\pi k(T_i - T_o)r_i r_o}{(r_o - r_i)}$$

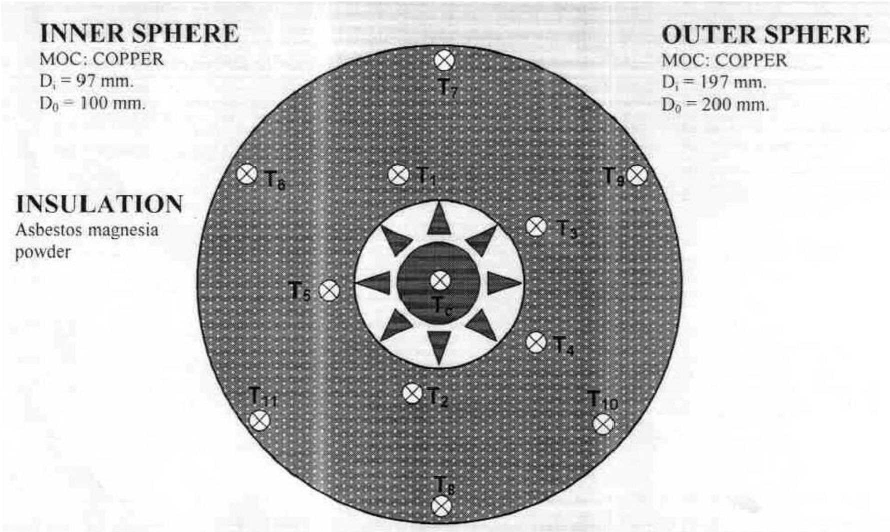
Where,

- r_i Inner sphere radius (m).
- r_o Outer sphere radius (m).
- T_i Temperature of the inner sphere surface (°C).
- T_o Temperature of the outer sphere surface (°C).

From the experimental values of q, T_i and T_o the unknown thermal conductivity K can be determined as

$$k = \frac{q * (r_o - r_i)}{4\pi (T_i - T_o) r_i r_o} \quad W / m\,°C$$

EXPERIMENTAL SET UP:



Thermal Conductivity of Insulating Powder

PROCEDURE:

1. Increase slowly the heater input by the dimerstat starting from ‘0’ position and this input must remain constant throughout the experiment.
2. The digital temperature controller will control the temperature near the set point value by switching on and off the heater.
3. Wait till a satisfactory steady state condition is reached. Reading temperatures of thermocouples 1 to 4 or 6 to 10 can check this.

OBSERVATIONS:

1. Outside diameter of the inner copper sphere(r_i) = 0.1 m
2. Inside diameter of the outer copper sphere (r_o) = 0.197 m
3. Heater Capacity = 450 Watt (aprrox.)
4. Insulation material = Asbestos magnesia powder

OBSERVATION TABLE:

Sr. No.	Voltmeter Reading, V (volt)	Ampere meter Reading, I (amp)	Heat Input q = V * I (watt)
1			
2			
3			

INNER SPHERE: TEMPERATURE INDICATED BY THERMOCOUPLES

Sr. No.	T_1	T_2	T_3	T_4	T_5	T_6	Mean Temp. $T_i = \frac{T_1 + T_2 + T_3 + T_4 + T_5 + T_6}{6}$
1							
2							
3							

OUTER SPHERE: TEMPERATURE INDICATED BY THERMOCOUPLES

Sr. No.	T_7	T_8	T_9	T_{10}	T_{11}	T_{12}	Mean Temp. $T_o = \frac{T_7 + T_8 + T_9 + T_{10} + T_{11} + T_{12}}{6}$
1							
2							
3							

CALCULATIONS:

$$k = \frac{q * (r_o - r_i)}{[4 * \prod * (r_i * r_o)(T_i - T_o)]} \text{ W / m °C}$$

RESULT:

Thermal conductivity of given insulating powder (k) = W / m °C

CONCLUSION:

REVIEW QUESTION:

1. List the assumption made for the experiment.
2. Describe the set-up and explain experimental procedure.

Marks obtained:	Signature of faculty:	Date:

EXPERIMENT NO 3
THERMAL CONDUCTIVITY OF METAL ROD

AIM:
To determine thermal conductivity of the metal rod at given section.

SPECIFICATIONS:

Length of the metal bar (Total)	= 500 mm.
Size of the metal bar (diameter)	= 25 mm.
Test length of the bar	= 300 mm.
No. of thermocouples mounted on the bar	= 6
Distance between two successive thermocouples	= 52 mm
Distance of first thermocouple from the heater	= 20 mm
No. of thermocouples in the insulation shell	= 4
Radial distance of thermocouple from the center of the rod	= 25 mm & 50 mm.
Axial Distance of thermocouple pairs embedded in the insulation from the heater end	= 100 mm & 200 mm
Heater coil (Bend Type)	= Nichrome heater.

Digital Temperature Indicator (0-200 °C with 0.1 °C Acc.) with Selector switch (12 Channel).

Positions 1 to 6	Thermocouple positions on metal bar.
Positions 7 to 10	Thermocouple positions in the insulating shell.
Positions 11 & 12	Thermocouple positions to measure the Cooling water inlet and outlet temperature.

Dimmerstat for heater coil	0 -240 V, 5 A
Voltmeter	0-250 V, Digital
Ammeter	0 -2 A, Analog
Measuring flask for water flow rate	1000 cc Volume.
Stop Watch	

Insulating Powder Asbestos (k = 0.03 W/ m °C)

THEORY:
The heater will heat the bar at its end and heat will be conducted through the bar to the other end.
After attaining the steady state,
Heat given away to the metal bar by heater (W),

$$Q_H = V * I$$

Heat flowing out of bar (i.e. the far end, at cooling jacket) (W).

$$Q_W = m * C_p * (T_{12} - T_{11})$$

Where,

m	Mass flow rate of cooling water (kg/sec)
C_p	Heat capacity of water (J / kg °C)
T_{11}	Inlet temperature of water (°C)

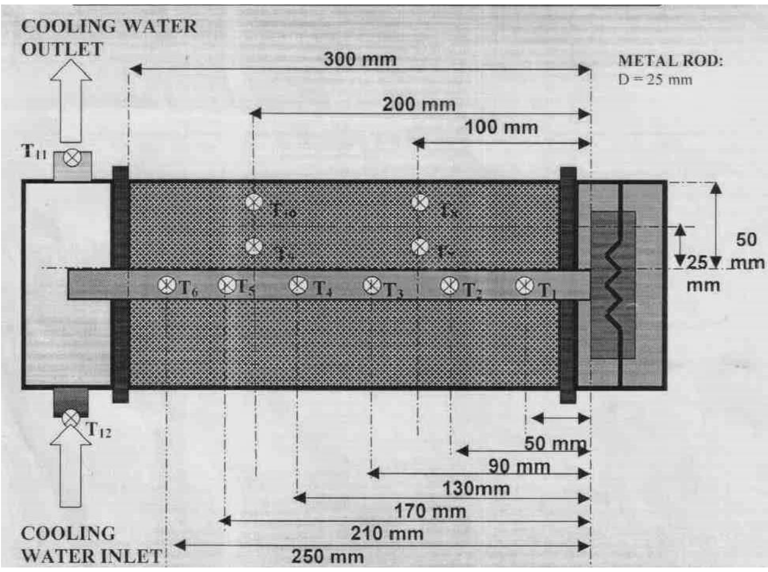
T_{12} Outlet temperature of water (°C)

Thermal conductivity of bar at any cross section AA can now be calculated as:

$$Q_{AA} = -k_{AA} * A * (\partial T / \partial x)_{AA}$$

Where,

Q_{AA}	Energy passing through the cross section AA
k_{AA}	Thermal Conductivity of metal bar at cross section AA.
A	Cross sectional area of the metal bar (m^2)
$(\partial T / \partial x)_{AA}$	Temperature distribution, in the bar along its length at cross section AA (obtained from experiment i.e. from the graph of Temperature vs Distance).



Thermal Conductivity of Metal Rod

The energy term Q_{AA} to be used for calculating the thermal conductivity at AA section must include the radial losses occurring in the radial direction.

Radial heat loss at point 1, 100 mm away from the heater can be calculated using the temperature values of T_7 and T_8.

$$Q_{loss,1} = \frac{[2 * \prod * k_{ins} * L * (T_7 - T_8)]}{\log_e (r_o / r_i)}$$

Where,

r_o = 50 mm, r_i = 25 mm, radial distance from the center of the rod

L Length of test section = 0.3 mm.

Similarly radial heat loss at point 2, 200 mm away from the heater can be calculated using the temperature values of T_9 and T_{10}.

$$Q_{loss,2} = \frac{[2 * \prod * k_{ins} * L * (T_9 - T_{10})]}{\log_e (r_o / r_i)}$$

Where,

r_o = 50 mm, r_i = 25 mm, radial distance from the center of the rod
L :Length of test section = 0.3 mm.

Average heat loss per unit length of the test section

$$Q_{loss\ avg} = \left(Q_{loss,1} + Q_{loss,2}\right)/2 \qquad \text{W/ m}$$

$$Q_{AA} = Q_H - \left(Q_{lossavg} * \left(X_{AA}/L\right)\right)$$

Where,

L	Length of test section metal rod, 0.3 m.
X	Distance of thermocouple from the heater or distance of section AA from heater (For example for T_1, X = 0.02 m, for T_2, X = 0.072 m, for T_3, X = 0.124 m, for T_4, X = 0.176 m, for T_5, X = 0.228 m and for T_6, X = 0.280 m)

PROCEDURE:

1. Make sure that the dimmerstat is at zero position before switching ON the heater.
2. Switch ON the main switch.
3. Start the cooling water supply through the jacket and adjust it about 50 - 100 cc per minute.
4. Give input to the heater by slowly rotating the dimmerstat and adjust it to current equal to 0.5 amps.
5. Go on checking the temperature at some specified time intervals say 5 minutes and continue this till a satisfactory steady state (i.e. No change in temperature with respect to time) condition is reached.
6. Note the mass flow rate of water in kg/ minutes and temperature rise in it (with the help of thermometers). And also note the temperature readings from T_1 to T_{12} from Temperature Indicator by using Selector Switch.
7. Repeat the procedure by decreasing the current value to 0.4, 0.3 and 0.2.

OBSERVATION TABLE:

Thermo-couple No.		**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Temp (°C)	**1**												
	2												
	3												
	4												

Sr. No.	Voltmeter Reading (V)	Ammeter reading (Amp)	Mass Flow Rate of Cooling Water (kg / min)	Inlet Temp. of Cooling Water (T_{11}) (°C)	Outlet Temp. of Cooling Water (T_{12}) (°C)
1					
2					
3					
4					

GRAPHS:

1. Plot the graph of the temperature distribution (at steady state) along the length of the metal bar using observed values (1) to (6) , for determining the slopes at section AA. Slope is nothing but $\left(\frac{dT}{dX}\right)$ at various desired points (X) on the plot of Temperature vs. Distance (T vs. X). Nature of the graph will be of straight line type / slightly curvature (Convex Downward).
2. Plot the graph of Thermal conductivity vs Temperature (1 to 6).

RESULTS AND DISCUSSION:

1) The temperature of the bar decreases along the length of the bar and can be plotted.
2) Thermal conductivity of two sections can be calculated & its variation with temperature can be studied.

CONCLUSION:

REVIEW QUESTION:

1. List the assumption made for the experiment.
2. Describe the set-up and explain experimental procedure.
3. Sketch the experiment set-up.

Marks obtained:	Signature of faculty:	Date:

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Heat and Mass Transfer (ME303) **Date:**

PRACTICAL NO 4
UNSTEADY STATE HEAT TRANSFER APPARATUS

AIM:
1) To study the unsteady state temperature response of finite geometric shapes.
2) To calculate the value of surface conductance (h).

APPARATUS:
Unsteady State Heat Transfer Apparatus Digital Temperature indicator with Selector Switch

APPARATUS DESCRIPTION:
The apparatus consist of a 20 litre SS 304 tank equipped with 1 kW heater and a drain valve. The tank is well insulated from outside to prevent heat losses to the surroundings. The temperature of the fluid inside the tank can be controlled by the regulator provided on the tank and is measured using the digital temperature indicator. Two shapes of known geometry and metals are provided with thermocouples to study the unsteady state heat transfer.

T_3 Copper Cylinder
T_4 Mild Steel Sphere
T_1 Hot Oil/ Water Bath Temperature
T_2 Cold Oil/ Water Bath Temperature

Shape 1:

Geometry	:	Cylinder
MOC	:	Copper
Diameter	:	50 mm
Height	:	150 mm
Density (ρ)	:	8900 kg/m^3
Specific Heat (C_p)	:	0.38 Kj/kg K
Area	:	$2\pi r^2+2\pi rh$

Shape 2:

Geometry	:	Sphere
MOC	:	Mild Steel
Diameter	:	50 mm
Density (ρ)	:	7830 kg/m^3
Specific Heat (C_p)	:	0.465 Kj/kg K
Area	:	$4\pi r^2$

PROCEDURE:
Fill the heating tank with fluid (water/ oil) and switch on the heater. Maintain the desired temperature of the water/ oil using the regulator provided on the tank. Note down this temperature (T_a) and initial temperature of the body as indicated by the ΔT. Now dip the selected body into this hot fluid with selector switch at proper thermocouple number and start

noting down the unsteady state temperature response (T). After the temperature reaches the steady state take the body out of the heating tank and note down the unsteady state temperature response of the body while cooling. Repeat the above procedure with second body.

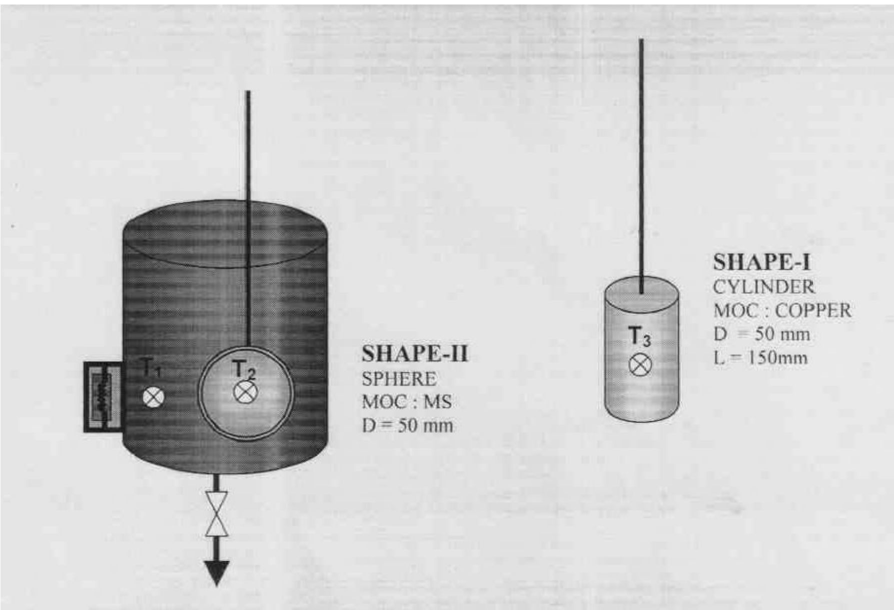
CALCULATION:

$$\frac{T - Ta}{Ti - Ta} = \exp\left\{-\frac{hA}{\rho Vc}\tau\right\}$$

$$\therefore h = \frac{\rho Vc}{A\tau}\log_e\left\{\frac{Ti - Ta}{T - ta}\right\}$$

Where,

C	=	Specific heat of body material
h	=	Surface conductance
A	=	Surface area of body
ρ	=	Density of material
K	=	Thermal conductivity of material
V	=	Volume of body
τ	=	Time in second
T_a	=	Fluid media temperature
T_i	=	Initial temperature of body



Unsteady State Heat Transfer Apparatus

OBSERVATION/RESULT TABLE:

Body Shape:

Cycle: **Heating**

Surrounding Temperature (Ta)=T_1 (heated water)=____________°C

Initial body Temperature (T_i)=T_3=____________°C for cylinder

T_4=___________°C for sphere

Sr. No.	**Time (τ) (sec)**	**Temperature Response Body temperature ($T=T_3$) (°C)**	**$\frac{(T - T_a)}{(T_i - T_a)}$**	Surface conductance **h (W/m^2-k)**	**Average h**
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
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30					

Body Shape:
Cycle: **Cooling**
Surrounding Temperature (Ta)=T_1 (heated water)=___________°C
Initial body Temperature (T_i)=T_3=__________°C for cylinder
T_4=___________°C for sphere

Sr. No.	**Time (t) (sec)**	**Temperature Response Body temperature ($T=T_3$) (°C)**	$\frac{(T - T_a)}{(T_i - T_a)}$	Surface conductance **h** **(W/m^2-k)**	**Average h**
1					
2					
3					
4					
5					
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Body Shape:
Cycle: **Heating**
Surrounding Temperature (Ta)=T_2(cold water)=___________°C
Initial body Temperature (T_i)=T_3=___________°C for cylinder
T_4=_____________°C for sphere

Sr. No.	**Time (τ) (sec)**	**Temperature Response Body temperature ($T=T_3$)°C**	**$\frac{(T - T_a)}{(T_i - T_a)}$**	Surface conductance **h (W/m^2-k)**	**Average h**
1					
2					
3					
4					
5					
6					
7					
8					
9					
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Body Shape:
Cycle: **Cooling**
Surrounding Temperature (Ta)= T_1 (heated water)=__________°C
Initial body Temperature (T_i)= T_3=_____________°C for cylinder
T_4=___________°C for sphere

Sr. No.	**Time (τ) (sec)**	**Temperature Response Body temperature ($T=T_3$)°C**	$\frac{(T - T_a)}{(T_i - T_a)}$	Surface conductance **h (W/m²-k)**	**Average h**
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
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30					

GRAPHS:

Plot $[(T-T_a)/(T_i-T_a)]$ vs. τ for heating as well as cooling cycle.

RESULT:

Average Surface conductance =

CONCLUSION:

REVIEW QUESTION:

1. List the assumption made for the experiment.
2. Describe the set-up and explain experimental procedure.
3. Sketch the experiment set-up.

Marks obtained:	**Signature of faculty:**	**Date:**

Heat and Mass Transfer (ME303) **Date:**

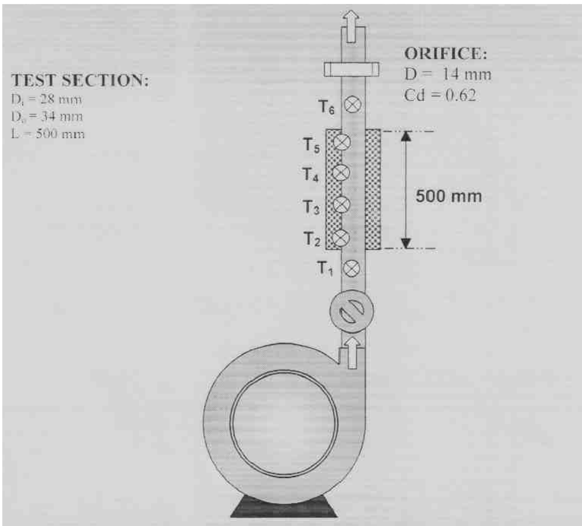
PRACTICAL NO 5
SPECIFIC HEAT OF AIR

AIM:
Determine the specific heat of air by forced convection.

APPARATUS:
Forced convection apparatus equipped with blower and manometer**.**

APPARATUS DESCRIPTION:
The apparatus consists of a blower unit fitted with the test pipe. Nichrome bend heater surrounds the test section. 4 Thermocouples are embedded on the test section and 2 thermocouples are placed in the air stream at the entrance and the exit of the test section to measure air temperatures. Test pipe is connected to the delivery side of the blower along with the orifice to measure the flow of air through the pipe. Input to the heater is given through dimmerstat and is measured by meter. It is also noted that only a part of the total heat supplied is utilized in heating the air. A temperature indicator with cold junction compensation is provided to measure the temperature in pipe wall in test section. Air flow is measured with the help of an orifice meter and the water manometer fitted on the board.

PROCEDURE:
- Start and adjust the flow by means of a valve to some desired difference in the manometer levels.
- Start the heating of the test section with the help of dimmerstat and adjust the desired heat input.
- Take the readings of all thermocouples at an interval of 10 minutes until the steady state is reached.
- Wait for steady state and take readings of all six thermocouples at steady state.
- Note down the heater input with the help of ammeter and voltmeter.



Forced convection apparatus

DATA:

Outer Diameter of the Pipe (D_o)	:-	34	mm
Inner Diameter of the Pipe (D_i)	:-	28	mm
Blower Motor Capacity	: -	1	H.P.
Orifice Diameter (do)	:-	14	mm
Coefficient of Discharge for Orifice	: -	0.64	

OBSERVATIONS TABLE 1:

Sr. No.	T_{1A} (^{0}C)	T_{6A} (^{0}C)	ΔT (^{0}C)	$T_{A\,AV}$ (^{0}C)	T_{2S} (^{0}C)	T_{3S} (^{0}C)	T_{4S} (^{0}C)	T_{5S} (^{0}C)	$T_{S\,AV}$ (^{0}C)
1									
2									
3									
4									

Where,

T_{1A}	=	Inlet temperature of air
T_{6A}	=	Outlet temperature of air
T_{2A} to T_{5A}	=	Temperature of tube (inner surface) at different location.
T_{6A} - T_{1A}	=	ΔT
$T_{A\,(AV)}$	=	$(T_{1A} + T_{6A})/2$
$T_{S\,(AV)}$	=	$(T_{2S} + T_{3S} + T_{4S} + T_{5S})/4$

CALCULATIONS:

(1) Pressure Differences

Δh_w = Pressure difference (Diff. in Manometer Limb $h_1 - h_2$) in terms of water

= ______ mm

Δh_A = Pressure difference in terms of air

= Δh_w x $(\rho_w - \rho_{air})/\rho_{air}$, density of air and water at $T_{A\,(AV)}$

= _______ mm
= _______ m

(2) Velocity of Air

$$Va = Cd \cdot \sqrt{\frac{2 \cdot g \cdot \Delta h_A}{1 - \frac{Ai}{Ao}}}$$

= _________ m / sec

(3) Volumetric Flow rate of Air,

Q_A = $A_i * V_a$
= _________ m^3/s

(4) Mass flow rate of air,

m = $Q_A * \rho_{air,TA}$
= _______ kg/s

(5) Rate Of Air Heating,

q_A = $m * C_p * \Delta T$, kcal / sec
But, q_A = $h * A_s * [T_{s, AV} - T_{a, AV}]$
Thus, Cp = $h * A_s * [T_{s, (AV)} - T_{a, (AV)}] / m * \Delta T$
= ___________ kcal / kg ^0C

Where,

C_d : Coefficient of discharge,
G : Acceleration due to gravity = 9.81 m / sec^2
H : Manometer reading, m
$\rho_{w,TA}$: Density of water at temperature at temperature $T_{A\ (AV)}$
ρ_{airTA} : Density of air at temperature at temperature $T_{A\ (AV)}$
Q_A : Volume of air-flow, m^3/ sec
Δh : Differential pressure expressed in meters of air
A_o : Cross-sectional area of orifice plate = $\Pi/4 * d_o^2$
A_i : Cross-sectional area of pipe = $\Pi/4 * d_i^2$
L : Length of test section.
C_p : Sp. heat of air at temperature $T_{A\ (AV)}$, kcal/kg ^0C
h : Heat transfer coefficient h_T at temperature $T_{A\ (AV)}$, kcal/hr.$m^2.^0C$

$$h = \frac{N_{ud} \times k_{air}}{d_i}$$

Where,

$$N_{ud} = 0.023(R_e)^{0.8}(P_r)^{0.4}$$

$$R_e = \frac{\rho v d}{\mu}$$

$$P_r = \frac{\mu C_p}{k_{air}}$$

A_s : Heat Transfer Area $= \Pi D_i L$

OBSERVATION TABLE 2:

Sr. No.	$\rho_{w,T}$ (kg/m^3)	$\rho_{air,T}$ (kg/m^3)	Δh_a (m)	V_a (m/s)	$Q_A = A * V_a$ (m^3/ sec)	$m = Q_A * \rho_{air,T}$ (kg / sec)

RESULT TABLE:

Sr. No.	m (kg / sec)	h (kcal/hr.m^2.^{0}C)	ΔT (^{0}C)	q_A (kcal / sec)	C_p (kcal / kg ^{0}C)

CONCLUSION:

REVIEW QUESTION:

1. List the assumption made for the experiment.
2. Describe the set-up and explain experimental procedure.
3. Sketch the experiment set-up.

Marks obtained:	**Signature of faculty:**	**Date:**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Heat and Mass Transfer (ME303) **Date:**

PRACTICAL NO 6
DOUBLE PIPE HEAT EXCHANGER

AIM:
To compare overall heat transfer coefficients obtained by operating the double pipe heat exchanger in co-current and counter current mode and compare it with theoretical values.

APPARATUS DESCRIPTION:
The apparatus consists of tube in tube type (concentric) heat exchanger with effective heat transfer length of 1.5 m. The hot fluid (hot water) obtained from an electric geyser flows through inner copper tube while the cold fluid (cold water) flows through the annulus between mild steel outer tube and copper inner tube.

The direction of flow of cold water can be altered (Counter/ Co-current flow) using the valves provided. If valve no. 1 and 3 are open while valve no. 2 and 4 are closed the unit operate as counter current flow double pipe heat exchanger and if valve no 2 and 4 are open while valve no. 1 and 3 are closed the unit operates as co-current double pipe heat exchanger.

CO-CURRENT FLOW

THERMOCOUPLE:
T_1 = C.W. INLET
T_2 = C.W. MIDDEL
T_3 = C.W. OUTLET
T_4 = H.W. INLET
T_5 = H.W. MIDDLE
T_6 = H.W. OUTLET

POSITION OF VALVES:
V_1 : OPEN
V_2 : CLOSE
V_3 : OPEN

COUNTER-CURRENT FLOW

THERMOCOUPLE:
T_1 = C.W. INLET
T_2 = C.W. MIDDEL
T_3 = C.W. OUTLET
T_4 = H.W. INLET
T_5 = H.W. MIDDLE
T_6 = H.W. OUTLET

POSITION OF VALVES:
V_1 : OPEN
V_2 : CLOSE
V_3 : OPEN

Inner Tube:

Material of Construction	: Copper tube
Inner Diameter (d_i)	: 15 mm
Outer Diameter (d_o)	: 18 mm
Effective Length (L)	: 1500x2 = 3000 mm
Mass Flow Rate	: 0.5 – 5 LPM

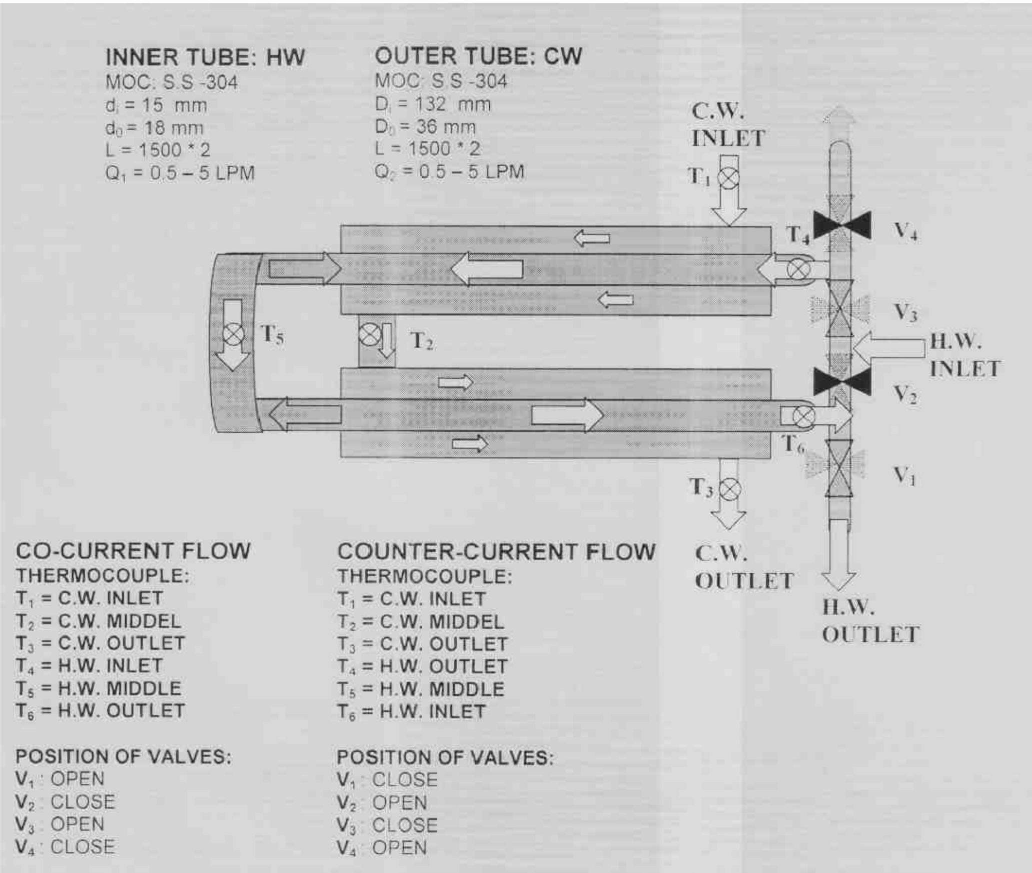
Outer Tube:

Material of Construction	: Mild Steel
Inner Diameter (D_i)	: 28 mm
Outer Diameter (D_o)	: 32 mm
Effective Length (L)	: 1500x2 = 3000 mm
Mass Flow Rate	: 0.5 – 5 LPM

The inlet and outlet temperatures of both the cold as well as hot water are measured by mercury in glass thermometers (0-110 °C) and flow rates by rota-meters (1-10 LPM, Fitzer make). Temperature and flow readings are recorded after the steady state is reached. The outer tube is provided with thermal insulation to protect heat transfer to the surroundings.

PROCEDURE:

- Place the thermometers in their respective positions and note down their readings when the equipments is at room temperature and no water is flowing in either side. This is required to correct the temperature measuring error.
- Start and set the flow of water on hot water side using the rotameter.
- After ascertaining constant flow of water from the exit end, switch ON the geyser.
- Start and set the flow of water on cold-water side using the rotameter. By operating the valves 1,2,3 and 4 as instructed to make the unit operate as either co-current or counter current heat exchanger.
- Keeping the flow rates same, wait till the steady state conditions are reached.
- Record the flow rates and temperatures at all the four places.
- Repeat the above, with the same value of flow rates and inlet temperature of hot water in another mode (counter/ co).
- Repeat the above experiment with different values of flow rates and temperatures.



Double pipe Heat Exchanger Setup

OBSERVATION TABLE:

Co-current Flow Run:

Sr. No.	Hot Water Side			Cold Water Side		
	Flow Rate m_h LPM	T_{hi} (°C)	T_{ho} (°C)	Flow Rate M_c LPM	T_{ci} (°C)	T_{co} (°C)
1						
2						
3						
4						

Counter current Flow Run:

Sr. No.	Hot Water Side			Cold Water Side		
	Flow Rate m_h LPM	T_{hi} (°C)	T_{ho} (°C)	Flow Rate M_c LPM	T_{ci} (°C)	T_{co} (°C)
1						
2						
3						
4						

CALCULATIONS:

For Co-current Flow Run:

Q_h = Rate of Heat Transfer from Hot Water
= $m_h * C_{ph} * (T_{hi} – T_{ho})$
=____________kcal/ hr
Q_c = Rate of Heat Transfer from Cold Water
= $m_c * C_{pc} * (T_{co} – T_{ci})$
=____________kcal/ hr
Q = $(Q_h + Q_c) / 2$
= ____________kcal/ hr
Where
C_{ph} = Specific heat of hot water at average temperature of T_{hi} & T_{ho}.
C_{pc} = Specific heat of cold water at average temperature of T_{hi} & T_{ho}.
Logarithm Means Temperature Difference as

$$LMTD = \frac{(T_{hi} - T_{ci}) - (T_{ho} - T_{co})}{\ln[(T_{hi} - T_{ci})/(T_{ho} - T_{co})]}$$

The overall heat transfer coefficient can be calculated using,

(1) Practical

For Inner tube side

$U_{overall}$ = Q/ (Ai * LMTD), kcal/ hr m^2 °C

For Outer tube side

$U_{overall}$ = Q/ (Ao * LMTD), kcal/ hr m^2 °C

Where, A_i = $\Pi d_i l$, $A_o = \Pi d_o l$.

(2) Theoretical

For Inner tube side

$$U_{overal} = \frac{1}{\frac{1}{\text{hi}} + \frac{di}{2k}\ln\frac{\text{do}}{\text{di}} + \frac{di}{do}\cdot\frac{1}{ho}}$$

For Outer tube side

$$U_{overal} = \frac{1}{\frac{1}{\text{ho}} + \frac{do}{2k}\ln\frac{\text{do}}{\text{di}} + \frac{do}{di}\cdot\frac{1}{hi}}$$

where,

h_i Heat transfer coefficient for inner (copper) tube flow

h_o Heat transfer coefficient for outer (mild steel)tube flow

k Thermal Conductivity of inner (copper) tube material

h_i is calculated by using the correlation $(N_u)_{di} = 0.023\ (Re)^{0.8}\ (Pr)^{0.3}$

$h_i\ d_i / k = 0.023 * (d_i\ m_h / \mu) * (C_p\ \mu/ k)^{0.3}$

Evaluate properties of water at average bulk mean temperature $(T_{hi}+T_{ho})/2$

h_o is calculated by using the correlation $(Nu)_{De} = 0.023\ (Re)^{0.8}\ (Pr)^{0.3}$

$h_o\ D_e / k = 0.023 * (D_e\ m_c / \mu) * (C_p\ \mu/ k)^{0.3}$

where D_e = Equivalent diameter of Annulus = (Di – do)

Evaluate properties at average bulk mean temperature $(T_{ci} + T_{co})/ 2$.

<u>For Counter current Flow Run:</u>

Q_h = Rate of Heat Transfer from Hot Water
= $m_h * C_{ph} * (T_{hi} - T_{ho})$
=____________kcal/ hr

Q_c = Rate of Heat Transfer from Cold Water
= $m_c * C_{pc} * (T_{co} - T_{ci})$
=_____________kcal/ hr

Q = $(Q_h + Q_c) / 2$
= ____________ kcal/ hr

Where

C_{ph} = Specific heat of hot water at average temperature of T_{hi} & T_{ho}.

C_{pc} = Specific heat of cold water at average temperature of T_{hi} & T_{ho}.

Logarithm Means Temperature Difference as

$$LMTD = \frac{(T_{hi} - T_{ci}) - (T_{ho} - T_{co})}{\ln[(T_{hi} - T_{ci})/(T_{ho} - T_{co})]}$$

The overall heat transfer coefficient can be calculated using,

(1) Practical

For Inner tube side

$U_{overall}$ = Q/ (A_i * LMTD), kcal/ hr m^2 °C

For Outer tube side

$U_{overall}$ = Q/ (A_o * LMTD), kcal/ hr m^2 °C

Where $A_i = \Pi d_i l$, $A_o = \Pi d_o l$.

(2) Theoretical

For Inner tube side

$$U_{overal} = \frac{1}{\frac{1}{\mathrm{hi}} + \frac{di}{2k}\ln\frac{\mathrm{do}}{\mathrm{di}} + \frac{di}{do}\cdot\frac{1}{ho}}$$

For Outer tube side

$$U_{overal} = \frac{1}{\frac{1}{\mathrm{ho}} + \frac{do}{2k}\ln\frac{\mathrm{do}}{\mathrm{di}} + \frac{do}{di}\cdot\frac{1}{hi}}$$

Where,

h_i	Heat transfer coefficient for inner (copper) tube flow
h_o	Heat transfer coefficient for outer (mild steel)tube flow
k	Thermal Conductivity of inner (copper) tube material

h_i is calculated by using the correlation $(N_u)_{di} = 0.023\ (Re)^{0.8}\ (Pr)^{0.3}$

$h_i\ d_i / k = 0.023 * (d_i\ m_h / \mu) * (C_p\ \mu/ k)^{0.3}$

Evaluate properties of water at average bulk mean temperature $(T_{hi}+T_{ho})/2$

h_o is calculated by using the correlation $(Nu)_{De} = 0.023\ (Re)^{0.8}\ (Pr)^{0.3}$

$h_o\ D_e / k = 0.023 * (D_e\ m_c / \mu) * (C_p\ \mu/ k)^{0.3}$

where D_e = Equivalent diameter of Annulus = (Di – do)

Evaluate properties at average bulk mean temperature $(T_{ci} + T_{co})/ 2$.

RESULT TABLE:

Sr. No.	COCURRENT FLOW				COUNTER CURRENT FLOW			
	Practical $U_{overall}$ kcal/ hr m^2 °C		Theoretical $U_{overall}$ kcal/ hr m^2 °C		Practical $U_{overall}$ kcal/ hr m^2 °C		Theoretical $U_{overall}$ kcal/ hr m^2 °C	
	Inner side	Outer side	Inner side	Outer side	Inner side	Outer side	Inner side	Outer side
1								
2								
3								

CONCLUSION:

REVIEW QUESTION:

1. List the assumption made for the experiment.
2. Describe the set-up and explain experimental procedure.
3. Sketch the experiment set-up.

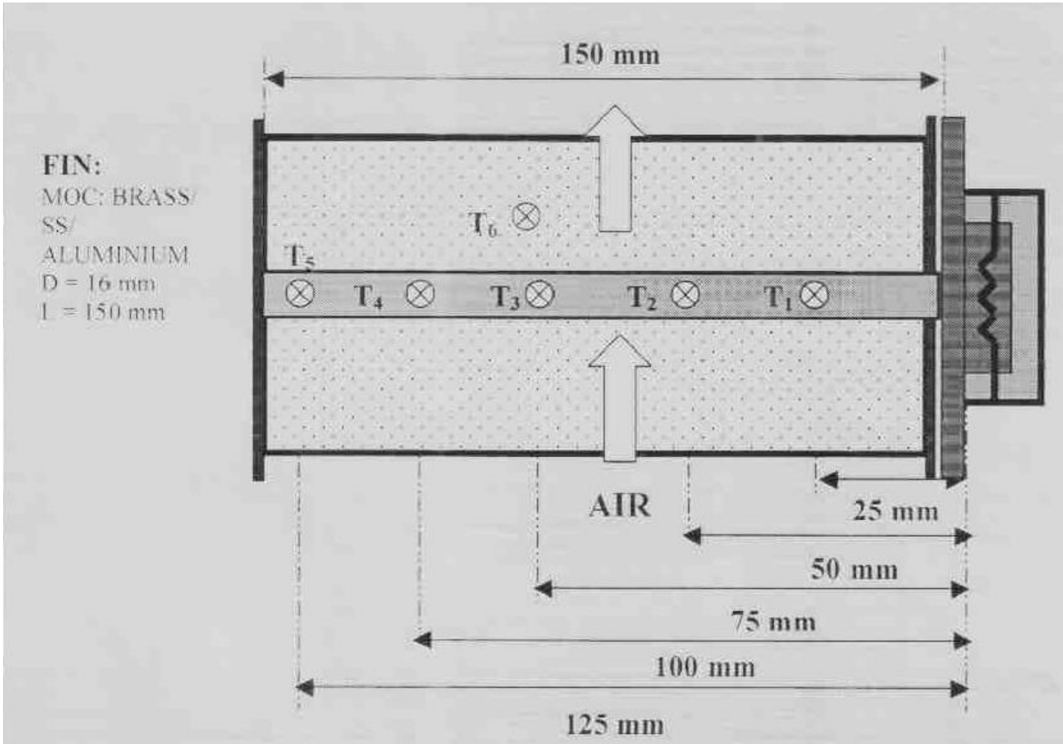
Marks obtained:	Signature of faculty:	Date:

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Heat and Mass Transfer (ME303) **Date:**

PRACTICAL NO 7
HEAT TRANSFER FROM PIN FIN

AIM:
To study the heat transfer from a pin fin under forced and natural convection and calculate the convective heat transfer co-efficient and effectiveness of the fin.



Heat Transfer from Pin Fin

SPECIFICATION:
Fins:
- Diameter 16 mm
- Effective length 150 mm
- No. of Thermocouple 5 Nos. positions along the length, made of Brass, Aluminum and Steel.
- Thermocouple located 25 mm away from each other
- First thermocouple at 25 mm from the base.

Heater:
- Capacity 350 W

Duct:
- Cross section Area 200 x 150 mm^2
- Length 1000 mm long connected to suction side of blower.

- Centrifugal Blower capacity 0.5 H.P. with orifice and flow control valve on discharge side.

Discharge line size:

- I.D. 50 mm
- O.D. 54 mm

Orifice:

- Diameter 25 mm
- Coefficient of discharge (C_d) 0.64

Measurements and controls:

- Dimmerstat to control heater input 0-230 V, 5 amps.
- Heater supply voltage (Digital) 0-250 V
- Heater current 0-2 amp.
- Twelve zone digital temperature indicator with selector switch.
- Water manometer connected to orifice-meter.

Thermal Conductivity of Fins:

- Brass fin 107 W/m °C.
- Aluminum fin 225 W/m °C.
- Stainless steel fin 16 W/m °C.

NOMENCLATURE:

T_m	Average fin temperature = $\frac{T_1+T_2+T_3+T_4+T_5}{5}$
ΔT	$T_m - T_f$
T_{mf}	Mean film temperature = $(T_m+T_f)/2$
ρ_a	Density of air, kg / m^3
ρ_w	Density of water = 1000 kg / m^3
d	Diameter of pin fin
d_o	Diameter of orifice
Cd	Coefficient of discharge of orifice = 0.64
μ	Dynamic viscosity of air, N-s / m^2
Cp	Specific heat of air, kJ / kg °C
ν	Kinematic viscosity, m^2 / s.
k_{air}	Thermal conductivity of air, w / m °C
β	Volume expansion coefficient = 1 / (T_{mf} + 273)
H	manometer difference, m of water
V	velocity of air in duct, m / s
Q	volume flow rate of air, m^3 / s
V_{tmf}	velocity of air at mean film temperature
A_c	Cross sectional area of fin

Note: All the properties are to be evaluated at Mean film temperature.

PROPERTIES OF AIR

T °C	ρ Kg/m³	C_p KJ/kg-K	$\mu \times 10^6$ N-Sec/m²	K W/m-K	Pr	$\nu \times 10^6$ m²/sec
0	1.293	1.005	17.2	0.0244	0.707	13.23
10	1.247	1.005	17.7	0.0251	0.705	14.16
20	1.205	1.005	18.1	0.0259	0.703	15.06
30	1.165	1.005	18.6	0.0267	0.701	16.00
40	1.128	1.005	19.1	0.0276	0.699	16.96
50	1.093	1.005	19.6	0.0283	0.698	17.95
60	1.060	1.005	20.1	0.0290	0.696	18.97
70	1.029	1.009	20.6	0.0297	0.694	20.02
80	1.000	1.009	21.1	0.0305	0.692	21.09
90	0.972	1.009	21.5	0.0313	0.690	22.10
100	0.946	1.009	21.9	0.0321	0.688	23.13
120	0.898	1.009	22.9	0.0334	0.686	25.45
140	0.854	1.013	23.7	0.0349	0.684	27.80

PART – I NATURAL CONVECTION:

PROCEDURE:

1. Open the duct cover over the fin. Ensure proper earthing to the unit and switch on the main supply.
2. Adjust dimmerstat to set the heater supply of around 80 volts
3. When the temperatures remain steady, note down the temperatures of the fin and duct fluid temperature.
4. Repeat the experiment at different inputs to heater.

OBSERVATION TABLE:

Sr. No.	Heater Input	Fin temperatures (°C)	Duct Temp (°C)

	V (Volt)	**I (Amp)**	**T_1**	**T_2**	**T_3**	**T_4**	**T_5**	**T_6**
1								
2								
3								
4								
5								

CALCULATIONS:

1. **Co-efficient for convective heat transfer, h (W/m°C)**

The fin under consideration is horizontal cylinder loosing heat by natural convection. For horizontal cylinder, Nusselt number,

Nu = 1.1 (Gr. Pr)$^{1/6}$ for $10^{-1} <$ **Gr. Pr** $< 10^{4}$
Nu = 0.53 (Gr. Pr)$^{1/4}$ for $10^{4} <$ **Gr. Pr** $< 10^{9}$
Nu = 0.13 (Gr. Pr)$^{1/3}$ for $10^{9} <$ **Gr. Pr** $< 10^{12}$

Where,

Gr = Grashof number, $\frac{g.\beta.d^3\Delta T}{\nu^2}$,

Pr = Prandtl number, $\frac{C_p\mu}{k_{air}}$

Determine, Nusselt number, Nu = $\frac{h.d}{k_{air}}$

$\therefore$ Co-efficient for convective heat transfer, h = __________.

2. **Rate of heat transfer (KW)**

$$Q = \sqrt{h.P.kAc}(T_m - T_f)\tanh ml,$$

where $$m = \sqrt{\frac{hp}{kAc}}$$

3. **Efficiency of the fin**

$$\eta = \frac{\tanh ml}{mL}$$

4. **Effectiveness of fin**

$$\varepsilon = \frac{Qactual}{Q\max}$$

$$= \frac{\sqrt{h.P.kAc}(T_m - T_f)\tanh ml}{hA_C(T_m - T_f)}$$

$$= \left(\sqrt{\frac{Pk}{hAc}}\right).\tanh ml$$

PART – II FORCED CONVECTION:

PROCEDURE:

1. Close the duct cover over the fin and start the bowler.
2. Adjust the dimmerstat so that about 100-110 volts are supplied to the heater.
3. After achieving the steady temperatures, note down all the temperatures and the manometer difference.

OBSERVATION TABLE:

CALCULATIONS:

As in natural convection, for horizontal cylinder loosing heat by forced convection.

$Nu = 0.615\ (Re)^{0.466}$ **for $40 < Re < 4000$**

$Nu = 0.174\ (Re)^{0.618}$ **for $40000 < Re < 40000$**

Where,

$$Re = \frac{V_{tmf}.d}{\nu}$$

Sr. No.	Heat Input		Manometer diff. (mm)	Fin temperatures (°C)					Duct Temp (°C)
	V	I		T_1	T_2	T_3	T_4	T_5	T_6
1									
2									
3									
4									

$$V_{tmf} = \frac{V * (T_{mf} + 273)}{(T_f + 273)}$$

Velocity of air is determined from air volume flow,

$$Q = C_d \frac{\pi . d_O^{\ 2}}{4} \sqrt{2.g.H(\rho_w - \rho_g)} \qquad m^3/sec$$

Where,
V = Velocity of fluid (m/s)

From Nusselt number, find out 'h' and from 'h' find out 'm'. Now temperature distribution, heat transfer rate, efficiency and effectiveness of the fin can be calculated using above equations.

OBSERVATIONS & RESULT TABLE FOR FORCED CONVECTION:

Sr. No.	Co-efficient of Convective Heat Transfer	Efficiency of fin	Effectiveness of fin
	h (W/m^2 K)	η	ε
1			
2			
3			
4			

CONCLUSION:

1) Comment on the observed temperature distribution and calculation by theory. It is expected that observed temperatures should be slightly less than their calculated values because of radiation and non-insulated tip.
2) Plot the graphs of temperature distribution in both natural and forced convection.

REVIEW QUESTION:

1. List the assumption made for the experiment.
2. Describe the set-up and explain experimental procedure.
3. Sketch the experiment set-up.

Marks obtained:	Signature of faculty:	Date:

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Heat and Mass Transfer (ME303) **Date:**

PRACTICAL NO 8
CRITICAL RADIUS OF INSULATION

AIM:
To study that heat transfer rate is maximum at critical radius of insulation.

SETUP:
The setup consists of following components.
Voltmeter(0-300V AC)
Ammeter(0-1 Amp AC)
2 four way temperature indicator(0-300^0 C)
Thermocouple(Type: Fe-, Range: 0-300^{0}C)
Cartridge heater(100Watts each)
Asbestos ropes for insulation

Pipe no	Inner diameter mm	Outer diameter mm	Radius of insulation mm
1.	24	28	14
2.	24	32	16
3.	24	36	18
4.	24	40	20

OBSERVATION TABLE:

Sr No	**Insulation surface temperature ^{0}C (T_I)**				**Heater surface temperature ^{0}C (T_H)**			
	T_1	**T_3**	**T_5**	**T_7**	**T_2**	**T_4**	**T_6**	**T8**

CALCULATIONS:

$$q_1 = \frac{t_i - t_o}{\frac{1}{2\pi L}\left[\frac{\log\frac{r_2}{r_1}}{K} + \frac{1}{r_2 * h_0}\right]}$$

t_1=reference temperature
t_2=ambient temperature
r_1=radius of heater=8mm
r_2=radius of insulation
L=length of heater=300mm
k=conductivity of insulation=0.949W/m^{o}k
h_0=heat transfer co-efficient of air=8.5W/m^2 ^{o}k

RESULT TABLE:

Sr No	Heat transfer rate in Watts			
	Q_1	Q_2	Q_3	Q_4

CONCLUSION:

Marks obtained:	Signature of faculty:	Date:

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Heat and Mass Transfer (ME303) **Date:**

PRACTICAL NO 9
STEPHAN BOLTZMANN LAW

AIM: To Measure The Value Of Stephan Boltzmann Constant.

SPECIFICATIONS:

Hemisphere dia.	0.205m
Base backelite plate	0.05m
Test disc dia	0.0195m
Thickness of test disc	0.003m
Thermocouples on test disc	1 nos
Water tank of capacity with immersion heater	8 litre
Themocouple on hemisphere	4nos
Temperature indicator	0-300 c

EXPERIMENTAL PROCEDURE

1. Start main switch of control panel
2. First heat the water in the water tank with the help of immersion heater up to boiling temperature. (T_6= 90-95^{0}C)
3. Then insert test disc in the backelite if not inserted (test disc is blackened totally)
4. Drop the boiled water on the hemisphere.
5. Take the readings of thermocouples on the hemisphere.
   i.e . T_1,T_2,T_3,T_4 up to steady state for only ones.
   Insert test disc below hemisphere.
   Take the readings of test disc. i.e. T_5 when it starts increasing with the help of the stop watch after every 10 sec. till the steady state is reached.

OBSERVATIONS:

Diameter of the test disc	d= 0.0195m
Mass of the test disc	m= 0.0076kg
Specific heat of test disc	Cp= 877 J/kg.K
Area of test disc	A_d= 2.98x10^{-4} m^2

TEMPERATURE OF HEMISPHERE:

Water temperature T_6	T_1	T_2	T3	T_4

TEMPERATURE OF TEST DISC:

Time in sec	10	20	30	40	50	60	70	80	90	100
T_5										
Time in sec	110	120	130	140	150	160	170	180	190	200
T_5										
Time in sec	210	220	230	240	250	260	270	280	290	300
T_5										
Time in sec	310	320	330	340	350	360	370	380	390	400
T_5										

CALCULATIONS:
Equation for Stephan Boltzmann constant

$$\sigma = \frac{\mathrm{m}\,\mathrm{C_p}\left(\frac{\mathrm{dT}}{\mathrm{dt}}\right)}{\mathrm{A_d}(T^4 - T_a^4)}$$

Where

m	Mass of test disc in kg	0.0076kg
C_p	Specific heat of test disc	877J/kg K
A_d	Area of test disc	$2.98\mathrm{x}10^{-4}\ m^2$
T	Mean temperature of hemisphere $T=(T_1 + T_2 + T_3 + T_4)/4$	
T_a	Average temperature of test disc. I.E. T_5	
dT	$T_{final}-T_{initial}$ T_{final}=steady state reading of test disc $T_{initial}$=temperature of test disc from which it starts increasing	
dt	Total time required for achieving the steady state from the time when temperature of the test disc starts increasing in Seconds.	

Heat and Mass Transfer (ME303) **Date:**

PRACTICAL NO 10
EMISSIVITY MEASUREMENT APPARATUS

AIM: To Measure The Emissivity Of The Yest Plate Surface At Various Temperatures.

SPECIFICATIONS:

1.	DIA OF TEST PLATE	0.175m
2.	Dia of black plate	0.175m
3.	Heater-Nichrome strip wound on mica sheet	
4.	Voltmeter	0-300 V
5.	Ammeter	0-5 A
6.	Enclosure	With one side perpex sheet
7.	Thermocouples	Chrome-Alumel-3nos
8.	Temperature indicator	0-300^{0}C
9.	Dimmerstate	

PROCEDURE:

1. Start the supply.
2. When both the toggle switches are in downward direction black platen dimmerstat operate where as when both the toggle switches are in upward direction test plate dimmerstat operate.
3. Gradually increase input to the heater to black plate and adjust the heater input to test plate slightly less then the black plate.
4. Check the temperatures of the two plates with small time intervals and adjust the input to test plate only by the dimmerstat so that the two plates will be maintained at the same temperature.
5. This some require some trail and error and one has to wait sufficiently to obtain the steady state condition.
6. after attaining the steady state condition record the temperature and voltmeter and ammeter readings for both plates.

OBSERVATIONS:

Temperatures	Steady state readings			
	1st time	2nd time	3rd time	4th time
Black plate(T_1)				
Test Plate(T_2)				
enclosure(T_3)				

CALCULATIONS:

$$\mathrm{W_a} - \mathrm{W_b} = \mathrm{A}(T^4 - T_a^4)(1 - E)\sigma$$

W_b	Heater input to black plate = V I watts
W_T	Heater input to test plate = V I watts
A	Surface area of plates
	d=0.175m
	t=0.006m
Ts	Temperature of plates after steady state(T_1=T_2)+273(^{0}K)
T_E	Enclosure temperature= T_3+273 ^{0}K
E	Emissivity of test plate which is to be obtained
σ	Stephan Boltzmann constant=5.667x10^{-8} W/m^2K

CONCLUSION:

Marks obtained:	Signature of faculty:	Date: